Is Spectral Centroid Enough?

Interpretable versus Differentiable Automatic Equalisation

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Project website: https://elena0129.github.io/elec5305-final-project-540995787/

Video demonstration: Recorded demonstration (4 min 43 s)

Working code and data: GitHub repository and run instructions

Abstract

Spectral centroid is a compact description of brightness, but it does not specify the frequency balance that an equaliser should restore. A controlled experiment compares a centroid-controlled high shelf, a three-band energy controller, and direct differentiable optimisation of three EQ gains. Thirty five-second GuitarSet excerpts from six performers were evaluated under eight known EQ conditions, giving 240 paired recovery tasks. Mean log-power spectral RMSE was 3.167138 dB for distorted inputs, 4.549816 dB for centroid correction, 0.017420 dB for the three-band controller, and 0.001708 dB for differentiable correction. Level-controlled and multi-resolution measurements supported the richer methods. A measured counterexample restored the centroid to within 0.000404 Hz while retaining 2.640119 dB of spectral shape error. Further experiments examined unmodified content, changed the objective while retaining one shelf, and varied corruption frequency and bandwidth. Differentiable correction had the lowest average error, but recovery declined under model mismatch. The findings support multi-band control as a practical compromise and separate the effects of target information, target choice, and EQ capacity.

Keywords: automatic equalisation; spectral centroid; differentiable DSP; parametric EQ; spectral matching; content dependence.

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1 Introduction

An automatic equaliser needs to decide both what should change and how a filter can produce that change. Spectral centroid offers an attractive starting point because it reduces a spectrum to one interpretable value. A lower centroid indicates that a larger share of the analysed power is concentrated at lower frequencies. However, that observation does not establish whether the signal has an unwanted EQ error. A different chord, register, or playing passage can change the centroid without any processing fault.

A concrete guitar example explains the difficulty. Suppose a recording is given a mid-frequency boost. A controller can adjust a high shelf until the average frequency returns to its original value. The high-shelf adjustment can move that average without removing the original mid-frequency emphasis. The numerical brightness target can therefore be met while the frequency balance remains different. This is a testable distinction between matching one descriptor and reconstructing a spectrum.

The central research question is: **How much information about tonal imbalance is retained by a single spectral centroid, and when do richer multi-band features or full-spectrum differentiable matching provide a useful improvement?** Two secondary questions concern whether a fixed target causes unnecessary changes to legitimate musical content, and whether good recovery depends on matching the corruption and correction filter structures.

Established equaliser structures [1], differentiable response matching [3], and context-dependent targets [5] motivate a controlled comparison of target information and filter capacity. The scope is static, reference-guided correction of short guitar excerpts: the original recording is an experimental reference, not a claim of ideal sound. The contribution is a common MATLAB comparison, measured-output audit, same-shelf objective ablation, and content and mismatch tests. These distinguish objective and processor limitations without proposing a new neural architecture.

2 Related work

2.1 Interpretable equalisation and its control targets

Välimäki and Reiss [1] review parametric and shelving equalisation as established ways to shape frequency response with a small number of meaningful controls. These structures are useful here because their effects can be inspected directly. Their interpretability does not determine how a target should be chosen: a filter gain is understandable, but a rule that chooses the gain can still be inappropriate for the input content. A single centroid and a small bank of band energies therefore provide useful, distinct control representations for the same recovery problem.

2.2 Differentiable DSP and parametric matching

DDSP integrates familiar signal-processing components with automatic differentiation [2]. Its original synthesis setting differs from EQ recovery, but the relevant principle is that meaningful DSP parameters can be adjusted through a differentiable model. Nercessian [3] applies differentiable biquads to parametric EQ matching and evaluates response errors rather than relying only on parameter agreement. This is important because similar parameter errors need not produce similar spectral errors. Differentiable means that derivatives of response error with respect to the gains can be computed and used to update those gains. It does not by itself imply a trained neural network.

Pepe et al. [4] optimise parametric IIR equalisation through a differentiable architecture, including filter frequency, gain, and quality factor, in acoustic equalisation settings. That work motivates response-based optimisation, but a room or car-cabin response is not the same target as the tonal balance of a musical excerpt. The present implementation uses the narrower problem of direct, per-excerpt gain optimisation. It adopts the differentiable response-matching principle, without training the neural networks or reproducing the complete experiments in those papers.

2.3 Context-dependent automatic EQ

Mockenhaupt et al. [5] identify the instrument in a track, select an instrument-specific spectral target, and predict parametric EQ settings from a spectral difference. The strength of this approach is that target selection is explicitly connected to context. Its target is a learned production profile, whereas the controlled recovery task here has the exact unmodified recording available. These are different assumptions. Restricting the experiment to guitar removes instrument-class variation, but it does not remove variation between guitar passages. The clean-content experiment examines that remaining issue.

2.4 Recent production systems and the selected scope

Sarkar and Lindborg [6] extend static parametric equalisation with per-band dynamic-range processing in a neural-driven multi-band processor. Their degradation-and-recovery training connects controlled corruption to practical automatic processing, but introduces learned inference and dynamics that are outside the current comparison. DiffVox [7] combines differentiable EQ, dynamics, delay, and reverberation for matching vocal effects. It demonstrates the value of retaining understandable effect parameters, while also showing how much broader production-style matching is than static EQ recovery.

These approaches motivate a smaller comparison in which the source of an improvement can be inspected: one descriptor, three regional powers, or a full spectral target. A centroid is inexpensive to store but ambiguous; band powers add interpretable regional control; a spectral objective retains more detail but requires a differentiable response and numerical fitting. The selected MATLAB study reproduces the response-matching principle [3,4] through direct gain optimisation. A same-shelf objective ablation then tests a small modification without changing processor capacity.

3 Methods

3.1 Recordings and controlled corruptions

GuitarSet provides guitar recordings with musical and performer metadata [8]. The formal evaluation used 30 mono five-second excerpts at 44.1 kHz: five source recordings from each of six performers, comprising 19 solo and 11 accompaniment excerpts. Three pilot sources were used for development and excluded from formal selection. Separation was by source recording, not performer; this does not test generalisation to unseen guitarists.

Reference preparation removed the excerpt mean and scaled its peak to 0.1. Source paths, excerpt positions, scaling information, and experiment settings were retained. A controlled EQ cascade then generated eight inputs for every reference. The filters were a 250 Hz low shelf, a 1000 Hz peaking section with 1000 Hz bandwidth, and a 2500 Hz high shelf. Shelves used slope 1. Table 1 defines the gain vectors in low, mid, high order.

Table 1. Known corruption gains for the 240 formal recovery pairs.

Condition	Low gain (dB)	Mid gain (dB)	High gain (dB)
Low cut	−6	0	0
Low boost	+6	0	0
Mid cut	0	−6	0
Mid boost	0	+6	0
High cut	0	0	−6
High boost	0	0	+6
Mixed A	+6	−6	+6
Mixed B	−6	+6	−6

All three automatic methods received the same distorted input and reference recording for a pair. The known corruption gains were used to generate inputs, not supplied as correction parameters. Each method produced one

static EQ setting per excerpt. This provides an offline comparison with a known target, rather than a blind or continuously adapting equaliser.

3.2 Shared spectral representation

Long-term power spectra were estimated using Welch averaging [9], with a periodic Hann window of 2048 samples, 1024-sample overlap, and a 2048-point FFT. Analysis retained frequencies from 20 to 16000 Hz. At 44.1 kHz, this gives 743 retained frequency values. Every method and metric used the same frequency grid within a pair.

For power values P_k at frequencies f_k, the power-weighted centroid is defined below. Here K is the number of retained bins; sums over k cover all K bins.

$$\mu(P) = [\Sigma_k f_k P_k] / [\Sigma_k P_k] \quad (1)$$

This definition is explicit because centroid values depend on spectral weighting and analysis settings. Multiplying the whole spectrum by a constant leaves the centroid unchanged. More importantly, many different normalised spectra can have the same centroid: fixing total power and one weighted mean does not determine all K spectral values. A small centroid error is consequently an objective achievement for Method A, but not sufficient evidence of tonal reconstruction.

3.3 The three automatic correction methods

Method A, Centroid, used one 2500 Hz high shelf. Its gain was constrained to ±12 dB. A coarse search in 0.5 dB steps was followed by a local search in 0.01 dB steps around the best coarse gain. Every candidate was applied to the distorted waveform and reanalysed. The selected gain minimised absolute reference-centroid error. This intentionally simple baseline can express a high-shelf correction, but cannot independently remove a mid-band peak or control low and high regions separately.

Method B, Three-band, compared integrated power in 20–500, 500–2000, and 2000–16000 Hz bands. It adjusted the three fixed EQ gains using signed band-power errors in dB. Starting at zero gain, the update was

$$e_b = 10 \log_{10}(E_{y,b} / E_{r,b})$$

$$\mathbf{g}^{(t+1)} = \mathrm{clip}_{[-12,12]} [\mathbf{g}^{(t)} - 0.7\, \mathbf{e}^{(t)}] \quad (2)$$

The controller retained the gains with the lowest summed squared band error, with at most 30 iterations and a 0.02 dB maximum-band-error stopping tolerance. Each candidate cascade was applied to the original distorted input. Band powers preserve regional information while remaining easy to interpret, although three integrated values cannot describe arbitrary detail inside a band.

Method C, Differentiable, used the same three nominal filter positions and optimised their gains against the full target spectrum. Its response model added the cascade's log-power response to the measured input spectrum:

$$g_s = 12 \tanh(\theta_s), \quad \omega_k = 2\pi f_k / f_{sample}$$

$$L_{y,pred}(f_k; \theta) = L_x(f_k) + \Sigma_s 10 \log_{10} |H_s(\exp(j\omega_k); g_s)|^2 \quad (3)$$

The objective was the mean squared difference between this prediction and the reference log-power spectrum. The gain variables started at zero. MATLAB automatic differentiation and Adam [10] used a learning rate of 0.03 for at most 400 iterations; optimisation could stop below 10^{-8} dB² predicted MSE. The best evaluated gains were retained. Only the three gains were optimised: frequencies, bandwidth, and shelf slope remained fixed. No neural network was trained. In (3), f_{sample} is the audio sample rate.

The differentiable response is an efficient fitting model, not the final evaluation signal. Selected gains were applied as causal IIR filters, corrected audio was saved as 24-bit WAV, and the saved waveform was read again for measurement. Finite values, sample dimensions, and clipping were checked. This separates predicted fitting accuracy from measured output accuracy.

3.4 Evaluation metrics and level control

Let r denote a reference and y a saved output. Define the log-power residual and raw RMSE as

$$d_k = 10 \log_{10}(P_{y,k} / P_{r,k})$$

$$R_{raw} = \sqrt{[(1/K) \Sigma_k d_k^2]} \quad (4)$$

Positive numerical floors were applied before logarithms. Raw RMSE measures both spectral shape and overall level agreement. To evaluate shape separately, the global level difference was removed using total power in the analysis band:

$$\Delta\ell = 10 \log_{10} [(\Sigma_k P_{y,k}) / (\Sigma_k P_{r,k})]$$

$$R_{shape} = \sqrt{[(1/K) \Sigma_k (d_k - \Delta\ell)^2]} \quad (5)$$

Shape RMSE is the level-controlled tonal metric; raw RMSE additionally checks absolute reference recovery. This controls analysis-band energy, not LUFS or perceptual loudness. Shape RMSE can exceed raw RMSE because the offset in (5) is set by total linear power, not by the mean residual that would minimise squared dB error. Absolute centroid error was $|\mu(P_y) - \mu(P_r)|$. Signed and absolute band-power errors were also retained, with the global level difference removed for their shape versions.

A separate multi-resolution log-magnitude RMSE used 512-, 2048-, and 8192-sample periodic-Hann STFTs with 75% overlap. Outputs were adjusted by the global level factor from (5). At each resolution, reference frames below −60 dB relative to peak frame energy were excluded. Magnitudes used a shared floor 80 dB below the reference peak magnitude. RMSE was computed over retained bins and frames, then averaged across the three resolutions. This is the stated evaluation metric, not a claim to reproduce every term of another library's multi-resolution loss.

Optimisation times excluded common file I/O and were summarised by their medians after warm-up in the main benchmark. Target-value count and optimised-gain count were recorded separately. A scalar feature need not produce the fastest implementation when its optimisation repeatedly filters and analyses audio.

3.5 Additional controlled experiments

Content dependence. The 30 unmodified clips were processed under their own centroid and a fixed target of 266.299647 Hz, the median of the three pilot centroids. The self target is a no-change control, not a learned predictor. A gain magnitude of at least 0.5 dB counted as an adjustment. Measurements included introduced spectral change, target error, and movement away from the clip's own unmodified centroid.

Matched-capacity ablation. A Spectrum-shelf variant replaced Method A's centroid objective with full-spectrum raw MSE while retaining the same high-shelf structure, ±12 dB limits, and coarse/fine search. Existing A, B, and C results supplied comparison points. The direct one-shelf comparison isolates the effect of changing the control objective and target information without adding filters. The three-filter methods provide context rather than another capacity-controlled comparison.

Model mismatch. A separate factorial test used peaking corruption at either 1000 or 1600 Hz, with bandwidth either 1000 or 300 Hz, and gain ±6 dB. The four configurations were matched, frequency mismatch, bandwidth mismatch, and joint mismatch. With 30 references, this produced another 240 pairs. The automatic correction models kept their original filter positions and bandwidth. An Oracle applied the known inverse corruption. It verifies recoverability with true parameters and is not an automatic competitor.

For each pair, recovery percentage was $100 \times [1 - R_{raw,method} / R_{raw,input}]$. Positive values indicate improvement; negative values indicate a larger error. Reported recovery percentages are averages of per-pair ratios, not ratios of the table's mean RMSE values. Mismatch effects were paired with the corresponding matched case using the same clip and gain sign.

3.6 Statistical and implementation checks

The 240 conditions share 30 references and six performers. They are not 240 independent recordings. Confidence intervals used 2000 bootstrap resamples of whole performer clusters, retaining clips, corruption conditions, and method pairings within a selected performer. The seed was 5305. The resulting intervals describe uncertainty within this small evaluation design, rather than broad population coverage.

MATLAB R2025b was used with Signal Processing Toolbox, Audio Toolbox, and Deep Learning Toolbox functions, including pwelch, designShelvingEQ, designParamEQ, dlarray, dlgradient, and adamupdate. New implementation concepts included differentiable biquad responses, separating shape from level error, and clustered resampling. Identity and pure-gain checks verified zero reference error and removal of a pure level change. Raw and centroid measurements were checked against the benchmark. Numerical outputs were saved before separate plotting, with checkpoints for completed fitting cases. Final MATLAB verification checked row counts, finite metrics, PNG files, and counterexample WAV-to-CSV agreement; the supplied report tables also agree after alignment by pair and method.

4 Results

4.1 Recovery accuracy and computation

All 240 formal pairs produced valid results for every method, with no failed rows. Table 2 summarises measured saved-audio performance. Method A substantially reduced mean centroid error, but increased mean raw spectral error relative to the distorted inputs. Thus, improvement in its intended descriptor did not imply improvement in the recovery task. Methods B and C reduced raw error for all 240 pairs. Method A reduced raw error in 36.25% of pairs and reached a gain boundary in 15.83%; B and C had no reported boundary hits.

Table 2. Main benchmark means over 240 pairs per method, except median fitting time. Lower errors and times are better.

Method	Centroid error (Hz)	Raw RMSE (dB)	Shape RMSE (dB)	MR RMSE (dB)	Fit time (s)
Distorted	163.159985	3.167138	3.361678	2.383855	—
A Centroid	7.882010	4.549816	3.685349	2.217727	1.8854
B Three-band	0.742357	0.017420	0.016942	0.012580	0.1978
C Differentiable	0.329024	0.001708	0.005018	0.004340	1.2441

The advantage remained after global level control and across STFT resolutions. C’s lower shape and multi-resolution errors therefore reflect more than matching overall energy. Its median fitting time was approximately 6.3 times B’s; A’s repeated waveform search was slower than both. A, B, and C used 1, 3, and 743 target values, respectively, and optimised 1, 3, and 3 gains. Target size, processor capacity, and fitting cost are different measures of complexity.

The raw-RMSE performer-cluster 95% intervals were [3.659013, 5.476906] dB for A, [0.015621, 0.018993] dB for B, and [0.001363, 0.002045] dB for C. The paired C-minus-B difference was −0.015713 dB, with interval [−0.017564, −0.013778] dB. This supports lower mean raw error for C in the tested matched setting. It does not establish an audible preference for such small residual differences.

Performance depended on the corruption. For high boost and high cut, A’s mean raw RMSE was 0.000000464 and 0.000099442 dB, respectively: its shelf structure can represent these inverse responses. A was more accurate than B and C in these two conditions. Its mean error instead reached 2.241389 dB for mid cut and 10.294172 dB for mixed B. Figure 1 shows the remaining regional errors after level control. A reduced mean absolute mid-band error only from 2.615830 to 2.364999 dB, compared with 0.012933 dB for B and 0.005803 dB for C. Thus, the aggregate ranking has clear conditional limits.

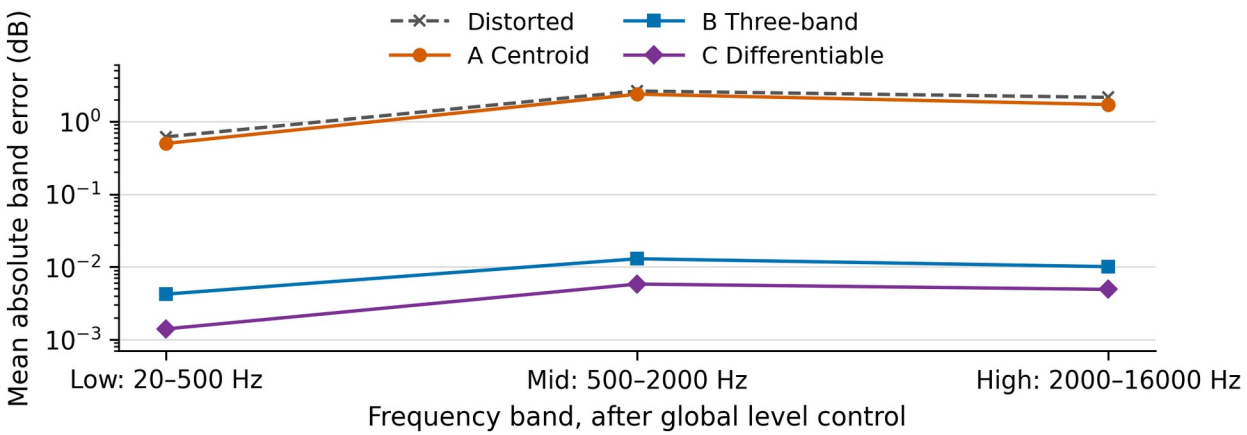


Figure 1. Mean absolute band-power error over 240 pairs per method after removing each output's global level difference. Lower values are better; the logarithmic axis shows both large residuals and the smaller errors of B and C. These are regional power errors, distinct from the spectral-bin RMSE in Table 2.

4.2 A measured counterexample

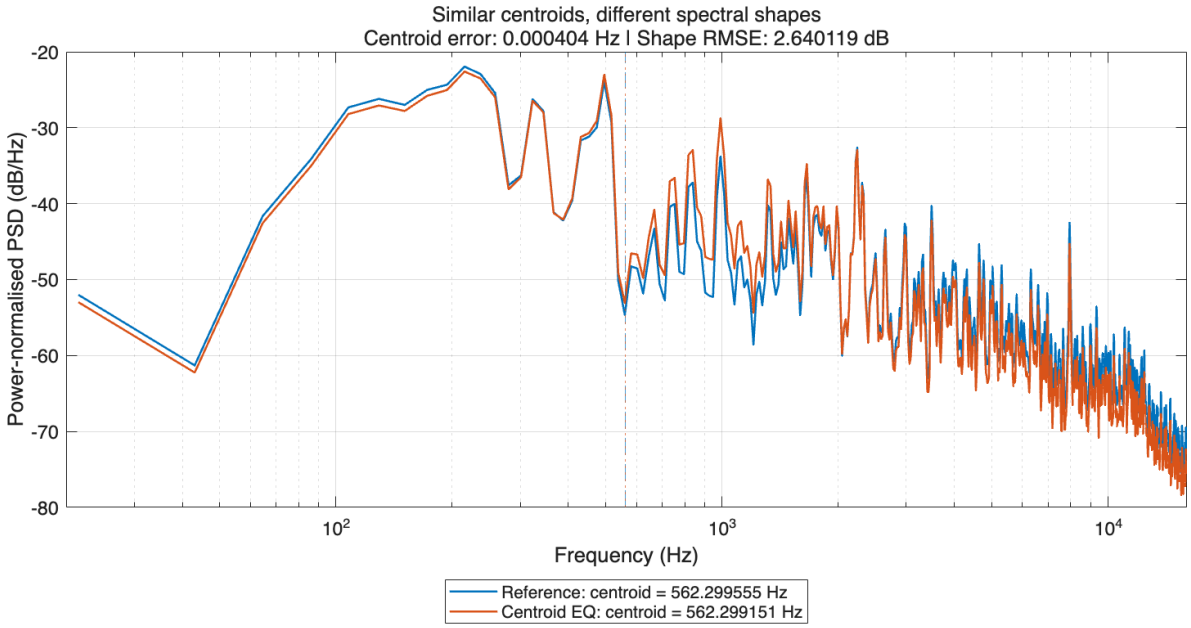


Figure 2. Formal case clip_005_mid_boost: power-normalised spectra after centroid correction. Reference and output centroids differ by only 0.000404 Hz, yet spectral shape RMSE is 2.640119 dB. The example illustrates a failure mode; aggregate results use every pair. The original MATLAB figure is shown.

The formal case clip_005_mid_boost (Figure 2) had a measured centroid error of 0.000403808 Hz, but shape RMSE of 2.640119 dB and raw RMSE of 2.012049 dB. The selected high-shelf gain was −1.90 dB. Among A's rows with centroid error below 1 Hz and shape RMSE above 1 dB, this case had the smallest centroid error. It is therefore a deliberately selected illustration, not an estimate of typical performance. MATLAB WAV remeasurement agreed with its saved values.

The example explains the scalar-target limitation in practical terms. A high shelf can move the power-weighted average back toward the reference while leaving the mid-frequency boost. Nearly coincident centroid markers therefore coexist with different spectral shapes. Matching the average frequency does not reconstruct the whole frequency distribution.

4.3 Dependence on unmodified musical content

The self-target control selected zero gain and introduced zero centroid and shape changes. With the fixed development target, every clean clip was adjusted by at least 0.5 dB. Mean absolute gain was 10.737667 dB, and 80% of clips reached a gain boundary. Mean target-centroid error decreased from 389.309216 to 173.207091 Hz, but the outputs moved an average of 216.108753 Hz away from their own unmodified centroids. Mean shape RMSE was 9.139064 dB, with performer-cluster interval [8.501769, 9.799086] dB.

These results distinguish target progress from preservation of content. The controller moved toward its specified target, yet changed recordings to which no artificial EQ error had been applied. The fixed target was not reached perfectly, and its large gains should not be described as successful tonal recovery. The result concerns this fixed target and task assumption; intentional production-style changes can have a different goal.

4.4 Changing the objective with the same shelf

Changing A's objective to full-spectrum MSE while retaining one high shelf lowered mean raw error by 3.693035 dB (95% interval [−4.629675, −2.802118] dB for Spectrum-shelf minus A). Raw error was lower in 75% of pairs, with 25% ties. Spectrum-shelf mean raw RMSE was 0.856782 dB; mean shape RMSE fell from 3.685349 to 1.722173 dB. Shape error improved in 55.83% of pairs, with 25% ties. The improvement was substantial but not universal.

This is evidence that changing the objective and target information can help even without adding filter controls. However, the remaining error was much larger than B's or C's. A richer target does not make a single high shelf capable of expressing every required correction. The experiment therefore identifies both objective-related and processor-related limitations without attributing the whole A-to-C difference to one cause.

Table 3. One-shelf objective comparison over the same 240 pairs. Processor structure, gain limits, and search procedure are held fixed.

Variant	Optimised gains	Raw RMSE (dB)	Shape RMSE (dB)
Centroid-shelf	1	4.549816	3.685349
Spectrum-shelf	1	0.856782	1.722173

4.5 Recovery under model mismatch

All 240 mismatch pairs completed, giving 1200 rows including the input and Oracle controls. Table 4 shows that C retained the lowest mean raw and shape errors among the three automatic methods in each scenario. Its mean raw error nevertheless increased from 0.000841 dB in the matched control to 0.870420, 0.378757, and 0.621835 dB under frequency, bandwidth, and joint mismatch. Mean recovery declined from 99.94% to 34.40%, 47.97%, and 14.43%, respectively.

Table 4. Model-mismatch mean raw RMSE in dB and Method C's mean per-pair RMSE reduction. Oracle uses true corruption parameters.

Scenario	Input	A	B	C	Oracle	C reduction (%)
Matched	1.327195	4.033572	0.020302	0.000841	0.000028	99.94
Frequency	1.326872	3.882311	1.753322	0.870420	0.000027	34.40
Bandwidth	0.727938	1.961926	0.611199	0.378757	0.000021	47.97
Joint	0.726666	1.622007	0.681080	0.621835	0.000022	14.43

B increased mean error in the frequency-mismatch condition, with mean recovery −32.14%. A had negative mean recovery in all four peaking conditions. The known inverse achieved very small, nonzero residuals; its displayed 100.00% recovery is rounded. It confirms that these synthetic corruptions can be reversed when the filter is known, rather than establishing an achievable automatic-method ranking.

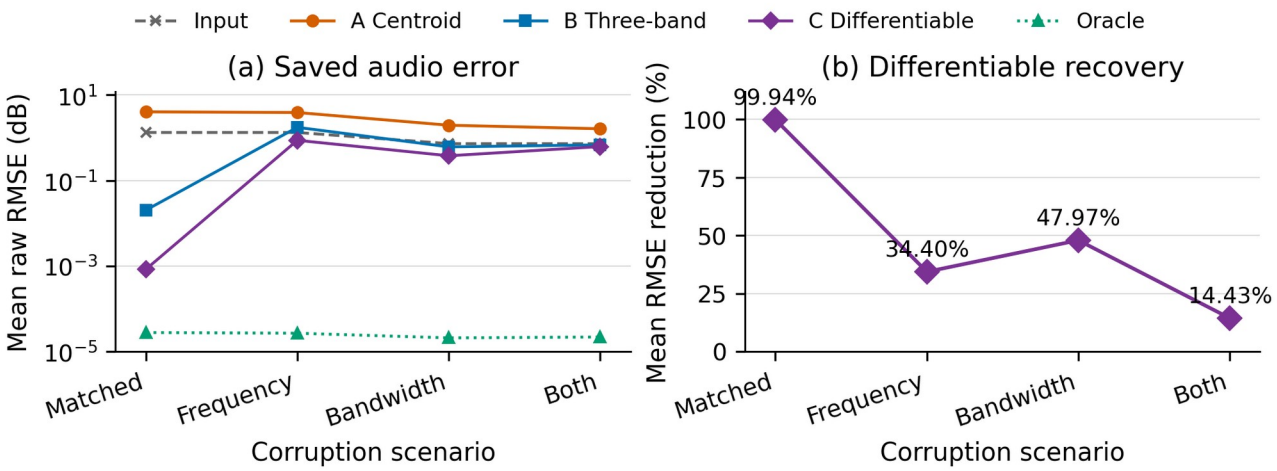


Figure 3. Recovery when the corruption centre and bandwidth differ from the fixed correction model. Panel a includes the known-inverse diagnostic; panel b shows Method C relative to each pair's own input error.

For C, paired mismatch-minus-matched raw error increases were 0.869579 dB for frequency mismatch, 0.377916 dB for bandwidth mismatch, and 0.620994 dB for joint mismatch. Their cluster intervals were [0.869460, 0.869731], [0.376494, 0.379643], and [0.620496, 0.621612] dB. All were positive. Joint mismatch need not have the largest absolute output error: narrowing the corruption also reduced the input's overall spectral error. Recovery percentages help make that change in starting difficulty visible.

5 Discussion

5.1 What the comparison establishes

The theoretical ambiguity of a weighted mean has a measured counterpart in the centroid counterexample. A can substantially improve centroid agreement while retaining a different spectral shape. B's much lower errors show that a small increase in regional information and correction controls can already recover most of the distortion in the matched benchmark. C gives the lowest numerical residuals by fitting the full spectral target, but this does not make the three methods interchangeable descriptions of the same information.

The main A-to-C comparison changes representation, objective, and filter freedom together. The one-shelf ablation addresses this criticism by holding the processor, bounds, and coarse/fine search procedure fixed. Its improvement supports the value of a spectrum-based objective within that shared structure. The remaining gap to B and C then shows why processor expressiveness also matters. It is still a comparison of two objectives, not proof that every use of a scalar feature must fail or that one specific objective is universally optimal.

5.2 Relation to modern approaches and engineering choices

The results are consistent with response-based differentiable matching [3,4]: optimisation should be judged by its output spectrum, rather than only parameter values or an indirect descriptor. The clean-content experiment supports the importance of contextual targets [5], without reproducing instrument classification or showing that an instrument-average target suits every passage. NDMP [6] adds learned inference and per-band dynamics; DiffVox [7] estimates parameters across a broader vocal-effects chain. Their tasks and evaluation sets differ from this static guitar benchmark, so their published performance cannot be ranked against these numbers.

The most accurate choice is not automatically the most useful engineering choice. B required about one-sixth of C's median fitting time and already had a mean shape error of 0.016942 dB. No controlled listening test establishes that listeners prefer C's 0.005018 dB residual. Conversely, A's one-dimensional target did not make its repeated waveform search fastest. Claims about low complexity should identify whether they concern feature storage, free DSP parameters, optimisation, or runtime filtering.

5.3 Limitations and alternative explanations

The main benchmark is favourable to B and C because corruptions use their nominal filter family. The mismatch test exposes a substantial loss of recovery when that assumption changes, but it samples only two centres, two bandwidths, and two gains. Oracle recovery shows that an exact inverse exists; it does not isolate how much automatic residual is caused by limited model capacity, local optimisation, finite iterations, or spectral-estimation effects. A broader parameter search and multiple initialisations would be needed to separate these explanations.

Only 30 excerpts and six performer clusters were evaluated, and source separation is not performer separation. Bootstrap intervals preserve the design's dependencies but do not replace broader sampling. Errors also depend on the chosen bandwidth, linear-frequency weighting, reference activity mask, and floors. Welch response prediction does not exactly describe short-record transient and windowing effects, which is one reason saved audio was remeasured.

Finally, energy normalisation controls a measurable global level difference without establishing equal perceived loudness. Magnitude-based recovery does not fully evaluate phase, temporal detail, or subjective production preference. Audio examples are informal demonstrations, not a formal listening experiment. The reference-guided task also provides target information unavailable to a blind practical equaliser.

5.4 What would be changed next

A follow-up should first expand recording and performer coverage, then test a preregistered set of corruption frequencies and bandwidths outside the correction model. Constrained optimisation of frequency and bandwidth, with several initialisations, would test whether added model flexibility explains the mismatch residuals. Log-frequency or perceptually weighted objectives could examine the effect of the current linear-frequency weighting. Context-conditioned targets should be assessed on clean passages as well as distorted ones, so that unnecessary intervention is measured directly. A controlled, loudness-matched listening study would be needed before making perceptual-quality claims. Dynamic adaptation would require separate tests of smoothing, update rate, and audible modulation.

6 Conclusion

Spectral centroid was insufficient as a general tonal-recovery target in this controlled guitar experiment. It could approach the reference centroid while preserving an incorrect spectral shape. A full-spectrum objective improved recovery even with the same single shelf, while the three-band and differentiable methods demonstrated the additional value of richer regional control. The three-band controller offered the strongest speed–accuracy compromise in the tested implementation, and differentiable matching produced the lowest mean spectral errors. Those very low matched-model residuals depended on the filter assumptions: recovery weakened substantially when corruption frequency or bandwidth changed. The defensible conclusion is therefore conditional. A simple centroid controller can suit a suitably matched shelf task, but broader tonal reconstruction requires an appropriate target, a sufficiently expressive processor, and evaluation beyond one brightness descriptor.

7 Reproducibility and demonstration

The saved studies contain 960 main-benchmark rows, 960 extended-evaluation rows, 60 clean-content rows, 960 capacity-ablation rows, and 1200 mismatch rows. Reused benchmark recordings are not independent observations. Source metadata, settings, gains and CSV tables provide the audit trail. The final local MATLAB R2025b check passed all nine regression tests, confirmed the expected rows and figures, remeasured the counterexample, and reached "Experiment verification complete."

Clone the repository including data/ and results/, and make its root MATLAB's Current Folder. Run `r = runtests('tests'); assert(~isempty(r) && all([r.Passed]));` followed by `run('run_project_demo.m')`. The supplied excerpts are sufficient; the raw dataset download is optional. The demonstration fits the pilot corrections and loads saved formal results. A fresh fit of all five formal modules instead uses `reproduce_results(false)`. The benchmark tables and fitting times reported here are recorded measurements, not a fresh full benchmark run in the

video. Omitted benchmark output WAVs can be reconstructed from recorded gains using restore_benchmark_audio("all"); reconstruction does not repeat optimisation.

The 4 min 43 s demonstration explains Methods A–C, shows MATLAB code and an actual run, reviews the results, and plays the reference, distorted input and three corrected pilot examples. The title-page links identify the project website, working repository and video. Sound examples illustrate the results rather than establish perceptual preference. The public dasp-pytorch automatic-EQ example was an implementation resource, distinct from the academic references below.

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